LITERATURE REVIEW

# Investment Decisions Data Platform

Context, Research Gap, Rationale, and Conceptual Framework

**PROJECT FOCUS**

The project aims to collect and organize financial-market and macroeconomic data for Apple Inc. (AAPL) through the Alpha Vantage API. The provided Python script requests historical daily stock prices, news sentiment, macroeconomic indicators, technical indicators, earnings data, and periodic latest-quote snapshots, then stores the responses in a JSON file for potential downstream analysis.

*Prepared for the Investment Decisions Data Platform project*

# 1. Establishing Context & Foundational Knowledge

Investment decisions are made in a changing environment influenced by company performance, market prices, investor sentiment, and broader economic conditions. A stock's price history can show how its market value has changed over time, while company earnings provide information about financial performance. News sentiment can help describe how financial news is characterized, and macroeconomic indicators—such as real GDP, consumer price index (CPI), the federal funds rate, Treasury yields, and unemployment—provide context about the wider economy.

The project's current data-collection script focuses on Apple Inc. (AAPL) and uses Alpha Vantage API endpoints to bring these different types of information into one local JSON file. It requests daily historical stock prices, news sentiment, earnings, and technical indicators including RSI, MACD, simple moving average (SMA), exponential moving average (EMA), and Bollinger Bands. It also requests selected economic series at their specified frequencies. Together, these datasets form a starting point for examining a company's market behavior alongside market news and economic conditions.

A key foundational consideration is that the data does not all share the same frequency or meaning. Stock prices and technical indicators are requested daily, while some economic indicators are requested monthly or annually. Before comparing them, the platform will need to preserve dates, understand the reporting frequency, and align the data carefully. The collected information can support analysis, but collecting it alone does not establish that a particular investment decision is correct.

# 2. Identifying the Research Gap

The project-specific gap addressed by the current implementation is the need for a single, organized data-collection layer that brings together several relevant input categories for investment research. In the supplied script, these categories are requested from separate API endpoints and saved under descriptive keys in investment_data.json. The script also periodically saves a latest-quote response with a retrieval timestamp.

At this stage, the implementation primarily retrieves and stores data. It does not yet provide a complete workflow for validating data quality, resolving missing or inconsistent observations, aligning daily market data with monthly or annual economic series, analyzing relationships among variables, comparing decision strategies, or evaluating whether any resulting signal would have been useful historically. These are practical development gaps between a data collector and a fuller investment-decision support platform.

**IMPORTANT SCOPE NOTE**

This is a preliminary, project-defined gap based on the supplied code and project scope—not a claim that no previous research or commercial platform has addressed these issues. A formal academic research gap should be confirmed by reviewing and citing relevant published studies and existing systems.

# 3. Justifying Your Research & Methodology

The project is justified by the practical value of organizing several types of investment-related data in a consistent place before analysis begins. Reviewing stock prices alone gives a limited view of the context in which prices change. Adding company earnings, news sentiment, technical indicators, and macroeconomic series creates a broader input dataset that can later be investigated systematically.

The current methodology is API-based data collection using Python. The requests library sends HTTP requests to Alpha Vantage; the returned JSON responses are checked for common API error messages and stored locally. The script uses a configuration dictionary to define the requested endpoints and parameters. It loads an existing JSON file when available, and repeatedly requests the GLOBAL_QUOTE endpoint, appending successful quote snapshots with a retrieval timestamp.

This approach is appropriate as an initial data-ingestion stage because it is straightforward to implement and makes the requested data categories explicit. However, the next stages should include endpoint-specific response validation, rate-limit handling, data cleaning, timestamp and frequency alignment, and reproducible analysis. If the project later produces investment scores or forecasts, the methods should be defined and evaluated separately rather than treating the raw API response as a recommendation.

## 4. Avoiding Duplication of Effort

Before adding advanced analysis, the project should review existing research and tools that combine stock-market data, technical indicators, news sentiment, and macroeconomic variables. This review can show which data sources, analytical methods, validation techniques, and evaluation measures are already commonly used. It can also help the team avoid rebuilding an existing method without a clear reason.

For this project, the most useful comparison is not simply whether another platform can retrieve stock prices. The team should examine how existing studies or systems combine different data frequencies, deal with missing observations, measure sentiment, test indicators, and evaluate decision methods. The project can then focus on a clear contribution, such as a transparent and maintainable collection pipeline, a reproducible data-preparation process, or a carefully evaluated analysis workflow.

A final literature review should record the source, method, data used, main findings, limitations, and relevance of each selected paper or system. Specific claims about previous studies should only be included after the sources have been read and verified.

## 5. Formulating a Theoretical or Conceptual Framework

The project can be organized around a simple conceptual framework: collect relevant data, validate and prepare it, combine observations on a consistent timeline, analyze the prepared data, and evaluate any resulting insights. The current code implements mainly the collection and initial storage stages; the later stages are proposed extensions.

| Stage | Inputs / activity | Purpose |
|---|---|---|
| **1. Data collection** | AAPL prices, news sentiment, earnings, technical indicators, and macroeconomic series from API endpoints | Gather potential explanatory information in one place |
| **2. Validation and storage** | Check response structure, API errors, missing values, dates, and file integrity; store structured records | Improve reliability and traceability |
| **3. Preparation and alignment** | Normalize fields and dates; handle daily, monthly, and annual frequencies appropriately | Make datasets suitable for comparison |
| **4. Analysis** | Explore price behavior and relationships with sentiment, technical indicators, earnings, and economic context | Generate evidence-based observations |
| **5. Evaluation** | Define metrics and test any proposed signal or strategy on appropriate historical data | Assess usefulness and limitations before drawing conclusions |

The main data groups in the supplied code are:

- Market data: daily AAPL historical prices and periodic GLOBAL_QUOTE snapshots.
- News data: NEWS_SENTIMENT results associated with AAPL.
- Macroeconomic data: real GDP, CPI, federal funds rate, 10-year Treasury yield, and unemployment.
- Technical indicators: RSI (14-period), MACD, SMA (20-period), EMA (20-period), and Bollinger Bands (20-period).
- Company fundamentals: AAPL earnings data.

Potential analysis variables include stock-price and return measures, technical-indicator values, sentiment scores, earnings observations, and macroeconomic values. The exact variables and analytical models should be selected only after inspecting the actual API response fields, data availability, and the project's research question.

## In Summary

The Investment Decisions Data Platform is currently a foundation for collecting and organizing financial and economic data, rather than a completed system that makes investment decisions. Its value is to prepare a broader and more traceable set of inputs for later analysis. A literature review will help establish the topic's background, verify the project's research gap, justify the chosen methodology, avoid unnecessary duplication, and guide the design of the analysis and evaluation stages.

## Suggested Literature-Review Search Topics

To turn this project-focused draft into a fully referenced academic literature review, search for published sources on:

- Financial decision-support systems and data integration for investment research.
- Use of technical indicators (RSI, MACD, moving averages, and Bollinger Bands) in financial analysis.
- Financial news sentiment analysis and its relationship to stock-market behavior.
- Relationships between macroeconomic indicators and equity markets.
- Financial time-series data quality, temporal alignment, and evaluation of forecasting or trading strategies.

**ACADEMIC INTEGRITY NOTE**

This document is tailored to the supplied project code and the five literature-review purposes provided. It does not invent paper titles, authors, statistics, or study findings. Add verified academic citations after selecting and reading relevant sources.